

Directions for Use

B. Braun Melsungen AG · 34209 Melsungen, Germany

Composition

10 ml of emulsion contain
Etomidate 20 mg

Excipients:

Soya bean oil, medium-chain triglycerides, glycerol, egg phospholipids for injection, sodium oleate, water for injections.

Pharmaceutical form

Emulsion for injection in glass ampoules of 10 ml

Pharmaco-therapeutic group

General anaesthetic

Indications

Induction of general anaesthesia

Contraindications

Etomidate-®Lipuro must not be administered to patients with known hypersensitivity to etomidate, soya bean oil or any other of the constituents of the product.

Etomidate has revealed porphyrogenic potential in animal experiments. Therefore it should not be administered to patients with inherited disorders of haem biosynthesis, unless the indication for administration of etomidate has definitely been established after careful consideration of its risks and potential benefits.

Newborns and infants up to the age of 6 months should be excluded from treatment with Etomidate-®Lipuro except for imperative indications during in-patient treatment.

Pregnancy, see section "Use in pregnancy and lactation" below.

Special warnings and precautions for use

Soya bean oil may rarely cause severe allergic reactions.

After prolonged continuous administration of etomidate there is a risk of transient adrenocortical failure.

Etomidate-®Lipuro may be used only by a doctor skilled in endotracheal intubation, with equipment for artificial respiration being available.

Etomidate-®Lipuro has no analgesic effect. If used for short-term narcosis, a strong analgesic, e. g. fentanyl, must be given prior to or simultaneously with Etomidate-®Lipuro; attention should also be paid to further information given under "Interactions".

Effects on the ability to drive or to use machines:

Even when Etomidate-®Lipuro is used as directed, patients having received this drug will not be able to drive or to use machines for at least 24 hours after administration.

Pregnancy and lactation

Safety of the use of Etomidate-®Lipuro during pregnancy has not been established. Therefore, Etomidate-®Lipuro should be administered to pregnant women only exceptionally, if there is no safer alternative.

Etomidate is secreted into breastmilk. If Etomidate-®Lipuro must be given during the lactation period, nursing is to be interrupted and not to be resumed before 24 hours after administration; breastmilk secreted during this period must be discarded.

Interactions

The hypnotic effect of etomidate is enhanced by neuroleptics, opioids, sedatives, and alcohol.

Etomidate-®Lipuro must not be mixed with other injection solutions without having previously been tested for compatibility.

Furthermore, Etomidate-®Lipuro must not be administered simultaneously with other injection solutions through the same line, unless compatibility has been established. Drugs to be given concurrently, e.g. an analgesic, must therefore be administered consecutively through the same line or through separate venous cannulae.

Etomidate-®Lipuro may be injected into the tubing of an infusion of isotonic sodium chloride having temporarily been stopped.

Dosage

The dosage is adjusted according to the individual response and the clinical effect.

The following dosage guidelines should be followed:

In general, the effective hypnotic dose is between 0.15 and 0.3 mg of etomidate per kg body weight, corresponding to 0.075 to 0.15 ml of Etomidate-®Lipuro per kg body weight.

Etomidate-®Lipuro

Children up to the age of 15 and elderly patients are given a single dose of 0.15 to 0.2 mg of etomidate, corresponding to 0.075 to 0.1 ml of Etomidate-®Lipuro per kg body weight. Also in patients belonging to these age groups, the exact dosage has to be adjusted according to the clinical effect.

In patients with liver cirrhosis and patients having been premedicated with neuroleptics the dose has to be reduced.

Method of administration

Etomidate-®Lipuro must be injected strictly intravenously and, as a rule, slowly (a single dose in approx. 30 sec) and in fractions if necessary.

Intra-arterial injection must be avoided as there is a danger of Etomidate-®Lipuro to cause necroses if injected intra-arterially. Paravenous injection will cause strong pain.

Prior to administration of Etomidate-®Lipuro appropriate premedication should be given in order to avoid the occurrence of myocloni. The use of benzodiazepines is recommended, e. g. diazepam which may be injected intramuscularly about 1 hour or intravenously 10 min. prior to administration of Etomidate-®Lipuro.

In patients with manifest epilepsy or with an increased tendency to convulsions, Etomidate-®Lipuro should be injected quickly, i. e. within a few seconds, in order to avoid too slow diffusion of etomidate into the brain. The good bioavailability of etomidate and its rapid distribution within the brain prevent activation of convulsions.

Overdose

In cases of overdose, especially if etomidate is combined with inhalation narcotics, the sleeping period may be extended and short periods of apnoea may occur.

When using Etomidate-®Lipuro, all equipment and medicaments usually required in general anaesthetic procedures should be available.

Undesirable effects

Etomidate inhibits the adrenocortical biosynthesis of steroids. After a single dose of etomidate the adrenocortical response to stressors is markedly reduced for approx. 4 - 6 hours.

After a single dose of etomidate, in unpremedicated patients, involuntary muscle movements (myocloni) are frequently observable. They correspond to the disinhibition of diencephalic excitations, similar to hypnogenic myocloni during physiological sleep. They can be prevented by premedication with opioids or benzodiazepines prior to the administration of etomidate.

Occasionally, after administration of etomidate, nausea and vomiting are observable, which are, however, caused predominantly by opioids given simultaneously or as premedication, further coughing, hiccough, and shivering.

Isolated cases of hypersensitivity reactions, up to anaphylactic reactions, have been reported. Yet, etomidate is the first-choice drug for patients with a history of allergy.

Soya bean oil may rarely cause severe allergic reactions.

There are isolated reports about the occurrence of laryngospasm after etomidate.

Especially after administration of higher doses of etomidate and if combined with central depressant drugs, transient apnoea may occasionally occur.

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Germany



Note
The doctor or pharmacist should be informed of any undesirable effect not mentioned in this leaflet.

Pharmacological Properties

Pharmacodynamic Properties

The drug substance etomidate acts as a hypnotic and is used in general anaesthesia.

Etomidate-®Lipuro is a pharmaceutical preparation in which the lipid-soluble etomidate is present in an emulsion of soya bean oil, medium-chain triglycerides and egg phospholipids for injection, not containing any solubilising agents. The vehicle emulsion has been used in parenteral nutrition for many years and is well tolerated.

Etomidate-®Lipuro allows a statistically significant reduction (p < 0.00001) of the local side effects observed for aqueous formulations.

The effect begins very rapidly, the duration of the hypnotic effect is short as a result of redistribution and degradation. A single dose of 0.3 mg/kg body weight leads within 10 s to unconsciousness and anaesthesia of 3–5 min duration.

The latency period for Etomidate-®Lipuro in EEG is 36 seconds, the start of the maximum effect is reached after 46 seconds.

Pharmacokinetic Properties

It is known from the literature that the half-life of the lipid particles is short. Etomidate rapidly separates from the oil particles on injection and this is expressed in the plasma concentration, which is comparable with that of the aqueous formulation. The disappearance of the lipid emulsion does not interfere with the half-life of etomidate.

Etomidate-®Lipuro and the aqueous formulation have comparable half-lives, distribution times and active times. Statistically significant differences could not be detected.

The plasma protein binding of etomidate (primarily to albumin) has been determined at 76.5%, it is reduced in renal dysfunction or chronic liver damage.

Cumulation has not been observed.

The total apparent distribution volume of 4.6 ± 2.2 l/kg is very large and points to considerable tissue uptake; the distribution volume in the central compartment of 24.2 ± 4.2 l or 21.2 ± 2.8 l or 23.2 ± 11.4 l is appreciably greater than the intravascular plasma volume, i.e. the central compartment also includes organs which are perfused by a large proportion of the heart minute volume (e.g. liver, brain), the redistribution from the deep peripheral into the central compartment is the rate-determining step for elimination.

The plasma concentration time curve has 3 phases, that can be fitted to an open 3-compartment model. Rapid distribution from the central compartment to a peripheral and a deeper peripheral compartment and a high elimination rate cause the plasma concentration to fall rapidly after a single administration (initial half-life 1.3 to 4.5 minutes). The large distribution volume ensures that the elimination half-life (terminal elimination half-life 2.4 – 5 h) is relatively long in spite of a high rate of hepatic extraction.

The lipophilic properties promote the rapid passage across the blood-brain barrier. The concentration in the cerebrospinal fluid approximately corresponds to the free, non protein-bound etomidate concentration (25 ± 3%).

The primary step of biotransformation is the hydrolysis of the ethyl ester in the liver. A small proportion is also subject to oxidative N-dealkylation. All metabolites discovered are pharmacologically inactive.

After the i.v. injection of 15 mg (radioactively labelled) etomidate base, ca. 75% of the total radioactivity appeared in the urine within 24 hours. Approx. 50% of the dose appeared in the first 4 hours after injection. Extrapolation allowed estimation that ca. 87% would be excreted in the urine after a sufficiently long period, i.e. other routes of excretion play a minor role. The major metabolite in the urine is the hydrolysis product of etomidate, namely R-(+)-1-(α-methylbenzyl)-5-imidazolecarboxylic acid, some of which is present as glucuronide and which accounts for approx. 80% of the radioactivity in the urine; with 55-59% in the form of the acid and 21–24% in the form of its glucuronide.

Preclinical safety data

Acute toxicity

LD₅₀ values are available for both etomidate solution and Etomidate-®Lipuro:

Species	LD ₅₀ (mg/kg body weight.)	ED ₅₀	Therapeutic ratio		
	S	E	(mg/kg body weight.)	S	E
Mouse	29.5	37.0	1.0	29	37
Rat	18.5	16.8	0.6	31	28
Guinea-pig	7.1	–	0.1	50	–
Rabbit	11.9	–	0.5	24	–
Dog	7.6	–	0.5	16	–

Legends
S = solution, E = Etomidate-®Lipuro,
Therapeutic ratio = LD₅₀/ED₅₀

Subchronic toxicity

ECG, blood pressure, haematological and biochemical analyses, urine investigations and histopathological investigations did not reveal any medicament-dependent side effects on daily administration of etomidate to rats over 3 weeks (highest dose 5.0 mg/kg body weight) and to dogs over 2 weeks (highest dose 1.50 mg/kg body weight).

Reproduction toxicology

Studies in rats and rabbits in which etomidate was administered in doses of 5 or 4.5 mg/kg body weight per day during the organogenic phase did not reveal any teratogenic or other embryotoxic effects. There have not been any investigations of administration during foetal development. Experience with administration to pregnant women is limited to the time of birth. Etomidate passes into the placenta when used for obstetric anaesthesia. The Apgar scores of the neonates are comparable with those after the administration of other hypnotics. The extent and clinical effects of steroid synthesis inhibition in the infantile adrenal cortex are not known.

Mutagenicity/carcinogenicity

Etomidate has not been tested for possible mutagenic effects. It is not possible in principle to exclude such an effect. Long-term studies of carcinogenicity in animals are not available.

Expiry date

The product must not be used beyond the expiry date stated on the labelling.

Instructions for storage / use / handling

Do not store above 25 °C.

Keep containers in the outer carton.

Etomidate-®Lipuro does not contain antimicrobial preservatives. After opening of the ampoule, the emulsion must be drawn up in a syringe under aseptic conditions and injected immediately, because fat emulsions promote microbial growth. Unused portions must be discarded.

Ampoules should be shaken prior to use to ensure homogenous distribution.

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Product Registration Holder:
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